ERYTHROMED GRANULES 200
(Erythromycin 200mg/5ml Granules for Oral Suspension)

Composition and Strength of Active Ingredients:
Each 5ml of reconstituted suspension contains Erythromycin Ethylsuccinate equivalent to Erythromycin 200mg

Product Description
An almost white powder when reconstituted forms a white, vanilla flavoured, homogenous suspension.

Pharmacodynamics

Erythromycin and other macrolides bind reversibly to the 50S subunit of the ribosome, resulting in blockage of the transpeptidation or translocation reactions, inhibition of protein synthesis, and hence inhibition of cell growth. Its action is predominantly bacteriostatic, but high concentrations are slowly bactericidal against the more sensitive strains. Because of the ready penetration of macrolides into white blood cells and macrophages there has been some interest in their potential synergy with host defence mechanism in vivo. Its actions are increased at moderately alkaline pH (up to about 8.5), particularly in Gram-negative species, probably because of the improved cellular penetration of the nonionised form of the drug.

Erythromycin has a broad spectrum activity. The pathogenic organisms which are usually sensitive to Erythromycin include:

a) Gram-positive cocci: *Streptococcus pneumoniae*, *Str. pyogenes. Staphylococcus aureus*.

b) Gram-positive organisms: *Bacillus anthracis*, *Corynebacterium diphtheriae* etc.

c) Gram-negative cocci: *Neisseria meningitidis, N. gonnorrhoeae, Moraxela catarrhalis*.

d) Gram-negative organisms: *Bordetella* spp., *Pasteurella, Haeemophilus ducreyi*. Among the gram-negative anacrobes more than half of all strains of *Bacteroides fragilis* and many *Fusobacterium* strains are resistant.

Fungi yeast, and viruses are resistant to erythromycin.

Pharmacokinetics
Facilitated absorption with empty stomach with maximum peak concentration after one hour of administration with elimination half life of approximately 2 hours. Doses may be given up to 4 times a day. Erythromycin ethylsuccinate exhibits lower extent of adverse effects due to the acidity produced by gastric juice as it absorbed from the small intestine with little metabolism and small amount, approximately 5% being excreted in the urine whilst majorly excreted by the liver and a significant proportion bound to serum proteins. Low concentration of Erythromycin ethylsuccinate was detected in the cerebrospinal fluid.

Indications

Indicated in the treatment of infections caused by susceptible strains of designated organisms in the diseases listed below: upper and lower respiratory tract infections that caused by *Streptococcus pyogenes* and *Streptococcus pneumoniae*. Pertussis (whooping cough) caused by *Bordetella pertussis*. Skin and soft tissues infections caused by *Streptococcus pyogenes* and *Streptococcus aureus*. Otitis media, erythrasma, listeriosis, diphtheria, primary syphilis in patients allergy to penicillin, prophylaxis of endocarditis.

Recommended Dose

For oral administration only.

Adults: 400mg (10ml), 6 hourly or 800mg (20ml), 12 hourly.

Dosage may be increased up to 4g per day according to the severity of the infection.

Children: 30-50mg/kg daily in divided doses; double in severe infections.

Children 2 to 8 years: 1g (25ml) daily in divided doses.

Infant and children up to 2 years: 500mg (12.5ml) daily in divided doses.

Route of Administration

Oral

Contraindications

Hypersensitivity to the active substance or to any of the excipients.

Erythromycin is contraindicated in patients taking simvastatin, tolterodine, mizolastine, amisulpride, astemizole, terfenadine, domperidone, cisapride or pimozide.

Erythromycin is contraindicated with ergotamine and dihydroergotamine.

Erythromycin should not be given to patients with a history of QT prolongation (congenital or documented acquired QT prolongation) or ventricular cardiac arrhythmia, including torsades de pointes.

Erythromycin should not be given to patients with electrolyte disturbances (hypokalaemia, hypomagnesaemia due to the risk of prolongation of QT interval).

Warning and Precautions

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)], ERYTHROMED GRANULES should be discontinued immediately and appropriate treatment should be urgently initiated.

There have been reports of infantile hypertrophic pyloric stenosis (IHPS) occurring in infants following erythromycin therapy. In one cohort of 157 newborns who were given erythromycin for pertussis prophylaxis, seven neonates (5%) developed symptoms of non-bilious vomiting or irritability with feeding and were subsequently diagnosed as having IHPS requiring surgical pyloromyotomy. Since erythromycin may be used in the treatment of conditions in infants which are associated with significant mortality or morbidity (such as pertussis or chlamydia), the benefit of erythromycin therapy needs to be weighed against the potential risk of developing IHPS. Parents and caregivers should be informed to contact their physician if vomiting and/ or irritability with feeding occurs.

This preparation contains Sodium metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites.

Erythromycin is excreted principally by the liver, so caution should be exercised in administering the antibiotic to patients with impaired hepatic function or concomitantly receiving potentially hepatotoxic agents. Hepatic dysfunction including increased liver enzymes and/or cholestatic hepatitis, with or without jaundice, has been infrequently reported with erythromycin.

Pseudomembranous colitis has been reported with nearly all antibacterial agents, including macrolides, and may range in severity from mild to life-threatening. Clostridium difficile-associated diarrhoea (CDAD) has been reported with use of nearly all antibacterial agents including erythromycin, and may range in severity from mild diarrhoea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon, which may lead to overgrowth of C. difficile. CDAD must be considered in all patients who present with diarrhoea following antibiotic use. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antibacterial agents.

Cardiovascular Events

Prolongation of the QT interval, reflecting effects on cardiac repolarisation imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in patients treated with macrolides including erythromycin. Fatalities had been reported.

Erythromycin should be used with caution in the following;

Patients with coronary artery disease, severe cardiac insufficiency, conduction disturbances or clinically relevant bradycardia.

Patients concomitantly taking other medicinal products associated with QT prolongation.

Elderly patients may be more susceptible to drug-associated effects on the QT interval.

There have been reports suggesting erythromycin does not reach the foetus in adequate concentrations to prevent congenital syphilis. Infants born to women treated during pregnancy with oral erythromycin for early syphilis should be treated with an appropriate penicillin regimen.

There have been reports that erythromycin may aggravate the weakness of patients with myasthenia gravis.

Erythromycin interferes with the fluorometric determination of urinary catecholamines.

Rhabdomyolysis with or without renal impairment has been reported in seriously ill patients receiving erythromycin concomitantly with statins.

Patients with rare hereditary problems of fructose intolerance, glucose-galactose malabsorption or sucrase-isomaltase insufficiency should not take this medicine.

To be taken into consideration by patients on a controlled sodium diet.

Interactions with Other Medicaments

Increases in serum concentrations of the following drugs metabolised by the cytochrome P450 system may occur when administered concurrently with erythromycin: acenocoumarol, alfentanil, astemizole, bromocriptine, carbamazepine, cilostazol, cyclosporin, digoxin, dihydroergotamine, disopyramide, ergotamine, hexobarbitone, methylprednisolone, midazolam, omeprazole, phenytoin, quinidine, rifabutin, sildenafil, tacrolimus, terfenadine, theophylline, triazolam, valproate, vinblastine, and antifungals e.g. fluconazole, ketoconazole and itraconazole. Appropriate monitoring should be undertaken and dosage should be adjusted as necessary. Particular care should be taken with medications known to prolong the QTc interval of the electrocardiogram.

Drugs that induce CYP3A4 (such as rifampicin, phenytoin, carbamazepine, phenobarbital, St John's Wort) may induce the metabolism of erythromycin. This may lead to sub-therapeutic levels of erythromycin and a decreased effect. The induction decreases gradually during two weeks after discontinued treatment with CYP3A4 inducers. Erythromycin should not be used during and two weeks after treatment with CYP3A4 inducers.

HMG-CoA Reductase Inhibitors: erythromycin has been reported to increase concentrations of HMG-CoA reductase inhibitors (e.g. lovastatin and simvastatin). Rare reports of rhabdomyolysis have been reported in patients taking these drugs concomitantly.

Contraceptives: some antibiotics may in rare cases decrease the effect of contraceptive pills by interfering with the bacterial hydrolysis of steroid conjugates in the intestine and thereby reabsorption of unconjugated steroid. As a result of this plasma levels of active steroid may decrease.

Antihistamine H1 antagonists: care should be taken in the coadministration of erythromycin with H1 antagonists such as terfenadine, astemizole and mizolastine due to the alteration of their metabolism by erythromycin.

Erythromycin significantly alters the metabolism of terfenadine, astemizole and pimozide when taken concomitantly. Rare cases of serious, potentially fatal, cardiovascular events including cardiac arrest, torsade de pointes and other ventricular arrhythmias have been observed.

Anti-bacterial agents: an in vitro antagonism exists between erythromycin and the bactericidal beta-lactam antibiotics (e.g. penicillin, cephalosporin). Erythromycin antagonises the action of clindamycin, lincomycin and chloramphenicol. The same applies for streptomycin, tetracyclines and colistin.

Protease inhibitors: in concomitant administration of erythromycin and protease inhibitors, an inhibition of the decomposition of erythromycin has been observed.

Oral anticoagulants: there have been reports of increased anticoagulant effects when erythromycin and oral anticoagulants (e.g. warfarin, rivaroxaban) are used concomitantly.

Triazolobenzodiazepines (such as triazolam and alprazolam) and related benzodiazepines: erythromycin has been reported to decrease the clearance of triazolam, midazolam, and related benzodiazepines, and thus may increase the pharmacological effect of these benzodiazepines.

Post-marketing reports indicate that co-administration of erythromycin with ergotamine or dihydroergotamine has been associated with acute ergot toxicity characterised by vasospasm and ischaemia of the central nervous system, extremities and other tissues.

Elevated cisapride levels have been reported in patients receiving erythromycin and cisapride concomitantly. This may result in QTc prolongation and cardiac arrhythmias including ventricular tachycardia, ventricular fibrillation and torsades de pointes. Similar effects have been observed with concomitant administration of pimozide and clarithromycin, another macrolide antibiotic.

Erythromycin use in patients who are receiving high doses of theophylline may be associated with an increase in serum theophylline levels and potential theophylline toxicity. In case of theophylline toxicity and/or elevated serum theophylline levels, the dose of theophylline should be reduced while the patient is receiving concomitant erythromycin therapy. There have been published reports suggesting when oral erythromycin is given concurrently with theophylline there is a significant decrease in erythromycin serum concentrations. This decrease could result in sub-therapeutic concentrations of erythromycin.

There have been post-marketing reports of colchicine toxicity with concomitant use of erythromycin and colchicine.

Hypotension, bradyarrhythmias and lactic acidosis have been observed in patients receiving concurrent verapamil, a calcium channel blocker.

Cimetidine may inhibit the metabolism of erythromycin which may lead to an increased plasma concentration.

Erythromycin has been reported to decrease the clearance of zopiclone and thus may increase the pharmacodynamic effects of this drug.

Pregnancy & Lactation

Pregnancy

There are no adequate and well-controlled studies in pregnant women.

Erythromycin has been reported to cross the placental barrier in humans, but foetal plasma levels are generally low.

There have been reports that maternal macrolide antibiotics exposure within 7 weeks of delivery may be associated with a higher risk of infantile hypertrophic pyloric stenosis (IHPS).

Lactation:

Erythromycin can be excreted into breast-milk. Caution should be exercised when administering erythromycin to lactating mothers due to reports of infantile hypertrophic pyloris stenosis in breast-fed infants.

Side Effects

Skin and Subcutaneous Tissue Disorders

Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP).

Postmarketing Experience:

Gastrointestinal Disorders: infantile hypertrophic pyloric stenosis.

Blood and lymphatic system disorders

Eosinophilia.

Immune system disorders

Allergic reactions ranging from urticaria and mild skin eruptions to anaphylaxis have occurred.

Psychiatric disorders

Hallucinations

Nervous system disorders

There have been isolated reports of transient central nervous system side effects including confusion, seizures and vertigo; however, a cause and effect relationship has not been established.

Eye disorders

Mitochondrial Optic Neuropathy

Ear and labyrinth disorders

Deafness, tinnitus

There have been isolated reports of reversible hearing loss occurring chiefly in patients with renal insufficiency or taking high doses.

Cardiac disorders

QTc interval prolongation, torsades de pointes, palpitations, and cardiac rhythm disorders including ventricular tachyarrhythmias.

Cardiac arrest, ventricular fibrillation (frequency not known).

Vascular disorders

Hypotension.

Gastrointestinal disorders

The most frequent side effects of oral erythromycin preparations are gastrointestinal and are dose-related. The following have been reported:

upper abdominal discomfort, nausea, vomiting, diarrhoea, pancreatitis, anorexia, infantile hypertrophic pyloric stenosis.

Pseudomembranous colitis has been rarely reported in association with erythromycin therapy.

Hepatobiliary disorders

Cholestatic hepatitis, jaundice, hepatic dysfunction, hepatomegaly, hepatic failure, hepatocellular hepatitis.

Skin and subcutaneous tissue disorders

Skin eruptions, pruritus, urticaria, exanthema, angioedema, Stevens-Johnson syndrome, toxic epidermal necrolysis, erythema multiforme.

Not known: acute generalised exanthematous pustulosis (AGEP).

Renal and urinary disorders

Interstitial nephritis

General disorders and administration site conditions

Chest pain, fever, malaise.

Investigations

Increased liver enzyme values.

Symptoms and Treatment of Overdose

In case of overdosage, erythromycin should be discontinued. Overdosage should be handled with the prompt elimination of unabsorbed drug and all other appropriate measures should be instituted.

Erythromycin is not removed by peritoneal dialysis or hemodialysis.

Effects on Ability to Drive and Use Machine

None known.

Mixing and Usage Instructions

Shake bottle until all granules flows freely, add boiled and cooled water to the 50ml mark on the bottle. Shake vigorously and then make up the volume to 60ml by adding more water to suspend the granules homogenously.

Shake well before use.

The concentration of erythromycin in the reconstituted suspension is 200mg/5ml.

Storage Condition

Before reconstitution, store below 30°C. Keep container tightly closed. Protect from heat and light.

After reconstitution, store below 30°C and discard any remaining suspension after 7 days.

Dosage Forms and Packaging Available

60 ml bottle

Manufactured by & Product Registration Holder:
KCK Pharmaceutical Industries Sdn. Bhd.
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